IN Phantom Dosimetry of Brachytherapy Source

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Medical Physics Laboratory II

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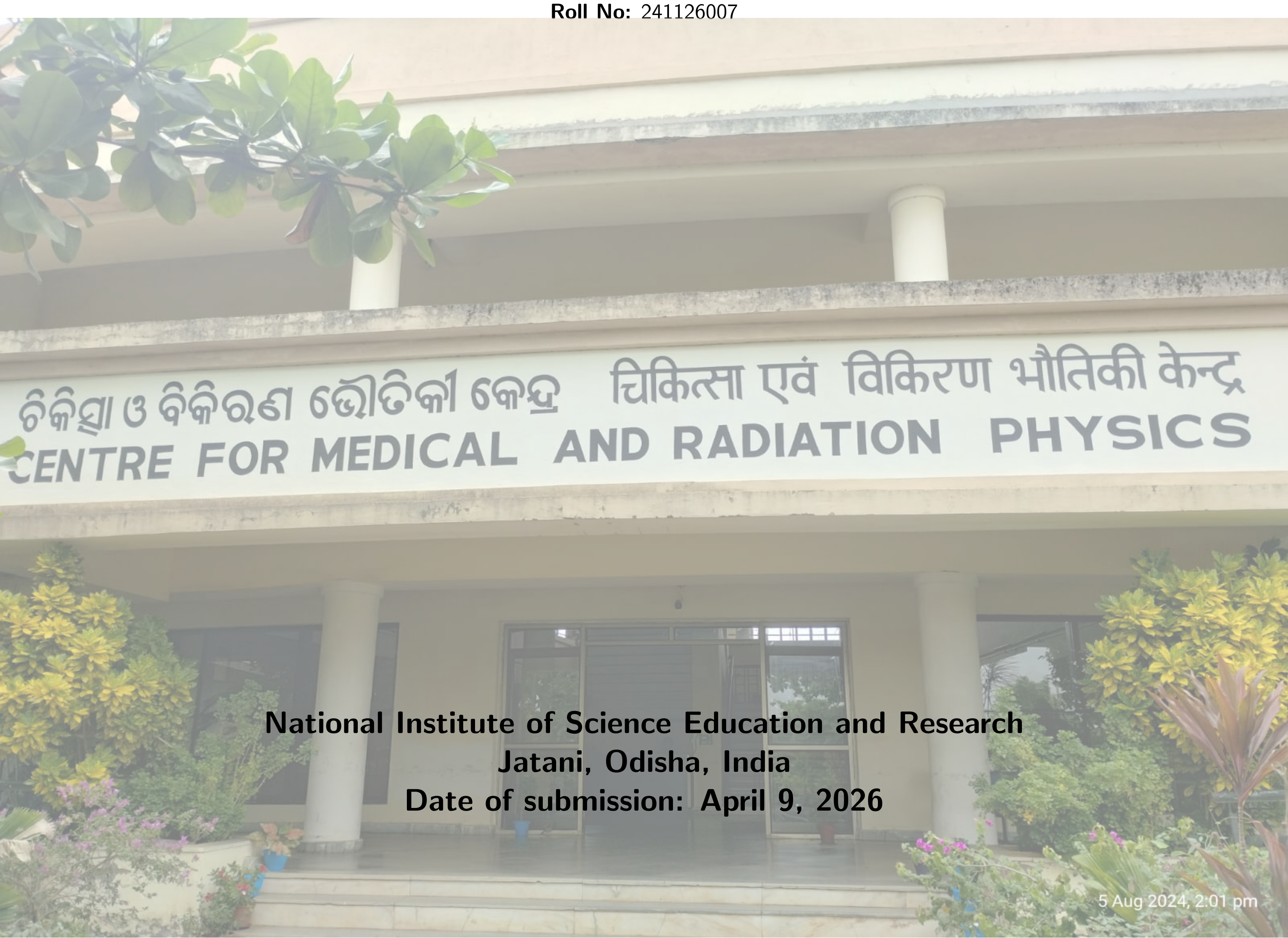

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1 Objective

The primary objectives of this experimental procedure are as follows:

1. To empirically determine and evaluate the fundamental TG-43 dosimetric parameters—specifically the Geometry Function $G(r, \theta)$, the Radial Dose Function $g(r)$, and the 2D Anisotropy Function $F(r, \theta)$—by executing in-phantom dosimetry on a clinical brachytherapy source.

2. To investigate and delineate the underlying physical and geometric factors that influence these dosimetric parameters.

2 Apparatus

The following specialized apparatus and instrumentation were employed during the execution of this experiment:

- **HDR Brachytherapy Unit:** Specifically, a Microselectron remote afterloading system housing a high-dose-rate Iridium-192 (^{192}Ir) therapeutic source.
- **Dosimetry Phantom:** A custom-designed, solid slab Perspex (IPD) phantom, dimensioned 30×30 cm^2 with a 10 mm thickness.
- **Source Delivery Mechanism:** A standard transfer tube securely compatible with interstitial plastic catheters.
- **Radiation Detectors:** Thermoluminescent Dosimeter (TLD) capsules, specifically doped Calcium Sulfate ($CaSO_4$:Dy).
- **Readout Device:** A specialized TL/OSL Research Reader for processing the irradiated dosimeters.

3 Theory

3.1 Challenges in Brachytherapy Dosimetry

Accurate dose assessment in clinical brachytherapy is intrinsically complex, governed by severe spatial constraints and extreme dose gradients. Measuring physical parameters surrounding miniature radioactive implants introduces complications relating to severe self-attenuation within the source matrix itself, highly asymmetric irradiation geometries, uncompensated tissue heterogeneities, and the profound manifestation of the inverse square law close to the radiation origin. High activity sources further exacerbate limitations associated with detector saturation and temporal resolution [1].

As a direct consequence of these empirical challenges, the standard paradigm for clinical brachytherapy heavily favors robust theoretical and computational models over routine in-vivo metrology. To mitigate the shortcomings of historical dose-calculation conventions—which often relied on overly simplistic approximations—the American Association of Physicists in Medicine (AAPM) promulgated the Task Group 43 (TG-43) report [2]. This formalism emerged as a universally accepted protocol due to its analytical clarity, inherent accuracy, and adaptability across various source configurations.

3.2 The TG-43 Dosimetry Formalism

The TG-43 formalism establishes a standardized methodology to compute the absorbed dose rate within a homogenous, tissue-equivalent medium. It segregates the physical phenomena dictating dose deposition—such as primary photon fluence, source construction effects, attenuation, and scatter—into independent, measurable, or calculable parameters [3].

Assuming cylindrical symmetry around the source's longitudinal axis, the 2D dose rate $\dot{D}(r, \theta)$ at an arbitrary spatial point $P(r, \theta)$ is mathematically described as:

$$\dot{D}(r, \theta) = S_K \cdot \Lambda \cdot \frac{G_L(r, \theta)}{G_L(r_0, \theta_0)} \cdot g_L(r) \cdot F(r, \theta) \tag{1}$$

Where:

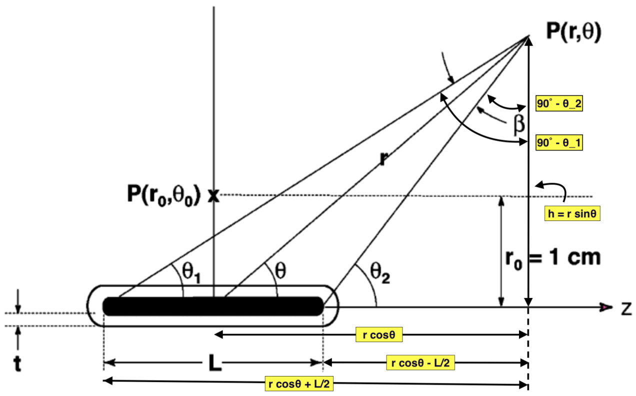

Figure 1: Schematic representation of the cylindrical coordinate system used in TG-43 dosimetry calculations. The source is located at the origin, with the active core aligned along the z-axis. The point of interest $P(r, \theta)$ is defined by its radial distance r and polar angle θ. The reference point $P(r_0, \theta_0)$ is located at $r_0 = 1$ cm and $\theta_0 = 90°$.

- r defines the radial distance from the geometric center of the active core to the measurement point P.
- r_0 is the universally adopted reference distance, fixed at 1 cm.
- θ corresponds to the polar angle subtended relative to the source's longitudinal axis.
- θ_0 is the reference angle, defining the transverse bisecting plane of the source, established at 90° (or $\pi/2$ radians).

3.3 Fundamental TG-43 Parameters

3.3.1 Air-Kerma Strength (S_K)

S_K characterizes the unattenuated output of the radioactive source and serves as the absolute scaling factor for the TG-43 equation. It is formally defined as the product of the air-kerma rate $\dot{K}_\delta(d)$ in vacuo and the square of the measurement distance d:

$$S_K = \dot{K}_\delta(d) \cdot d^2 \tag{2}$$

S_K is expressed in units of μGy m^2 h^{-1} (commonly denoted as U). The measurement of $\dot{K}_\delta(d)$ occurs in free space on the transverse plane at a macroscopic distance (typically 1 meter) where the source appears as a point. The threshold parameter δ deliberately suppresses low-energy contaminant photons (e.g., Titanium characteristic X-rays under 5 keV) which marginally inflate the in-air kerma without contributing biologically relevant dose in tissue [2]. The "in vacuo" specification demands rigorous corrections for ambient air attenuation, systemic scatter, and room geometry effects.

3.3.2 Dose Rate Constant (Λ)

The Dose Rate Constant acts as the conversion metric bridging the physical source output (S_K in air) and the absorbed dose in the medium at the specific reference locus $P(r_0, \theta_0)$:

$$\Lambda = \frac{\dot{D}(r_0, \theta_0)}{S_K} \tag{3}$$

Λ is intrinsically linked to the specific radionuclide employed and the intricate internal architecture of the source capsule.

3.3.3 Geometry Function ($G(r,\theta)$)

The geometry function isolates and quantifies the spatial attenuation strictly dictated by the inverse square law, purposely ignoring phenomena related to tissue absorption or scattering. Depending on the volumetric distribution of the active material, it is modeled either as a theoretical point or a 1D line:

$$G_P(r,\theta) = r^{-2} \quad \text{(Point Source Approximation)} \tag{4}$$

$$G_L(r,\theta) = \begin{cases} \frac{\beta}{Lr\sin\theta} & \text{if } \theta \neq 0^\circ \\ (r^2 - \frac{L^2}{4})^{-1} & \text{if } \theta = 0^\circ \end{cases} \quad \text{(Line Source Approximation)} \tag{5}$$

Here, L is the physical active length of the isotope, and β represents the angle (in radians) geometrically bounded by the extremities of the active source to the point $P(r,\theta)$.

For a line source centered at the origin, let θ_1 and θ_2 be the angles subtended by the two ends of the active length at the point $P(r,\theta)$. Then,

$$\beta = \theta_2 - \theta_1 \tag{6}$$

where

$$\tan\theta_2 = \frac{r\sin\theta}{r\cos\theta - L/2} \tag{7}$$

$$\tan\theta_1 = \frac{r\sin\theta}{r\cos\theta + L/2} \tag{8}$$

Therefore, the angular span is given by

$$\beta = \tan^{-1}\left(\frac{r\sin\theta}{r\cos\theta - L/2}\right) - \tan^{-1}\left(\frac{r\sin\theta}{r\cos\theta + L/2}\right) \tag{9}$$

3.3.4 Radial Dose Function ($g(r)$)

The radial dose function maps the variation in dose rate solely attributable to photon absorption and bulk tissue scattering strictly along the transverse plane ($\theta_0 = 90^\circ$). It specifically divides out the purely spatial divergence accounted for by the geometry function:

$$g_X(r) = \frac{\dot{D}(r,\theta_0)}{\dot{D}(r_0,\theta_0)} \cdot \frac{G_X(r_0,\theta_0)}{G_X(r,\theta_0)} \tag{10}$$

3.3.5 2D Anisotropy Function ($F(r,\theta)$)

The anisotropy function $F(r,\theta)$ characterizes the angular distortion of the dose distribution relative to the transverse axis. This deviation from isotropy arises largely from self-absorption within the heavy-metal active core and the oblique filtration pathway through the protective titanium or steel encapsulation at steep polar angles:

$$F(r,\theta) = \frac{\dot{D}(r,\theta)}{\dot{D}(r,\theta_0)} \cdot \frac{G_L(r,\theta_0)}{G_L(r,\theta)} \tag{11}$$

3.4 Experimental Derivation using TLDs

While $G_X(r,\theta)$ is a purely mathematical construct derived from physical source dimensions, $g(r)$ and $F(r,\theta)$ mandate empirical validation. These metrics are proportional ratios of absolute dose rates at distinct spatial coordinates. In a controlled experimental setup using solid-state dosimeters such as TLDs, provided that the physical configuration and irradiation parameters remain uniform, the accumulated TL signal (counts) is directly proportional to the absorbed dose rate. By executing ratio-based calculations, the absolute calibration constants of the TLDs mathematically cancel out, allowing the extraction of the TG-43 parameters using pure raw counts [3]:

$$g_X(r) = \frac{\text{TLD}(r,\theta_0)}{\text{TLD}(r_0,\theta_0)} \cdot \frac{G_X(r_0,\theta_0)}{G_X(r,\theta_0)} \tag{12}$$

$$F(r,\theta) = \frac{\text{TLD}(r,\theta)}{\text{TLD}(r,\theta_0)} \cdot \frac{G_L(r,\theta_0)}{G_L(r,\theta)} \tag{13}$$

3.5 Phantom Design and Methodology

Measurements are conducted utilizing a bespoke "IPD" Perspex phantom, a tissue-equivalent slab measuring $30 \times 30 \times 1$ cm^3. The center features a 2 mm diameter grooved channel specifically milled to accommodate an interstitial brachytherapy catheter. Arrayed along concentric arcs radiating from this central axis are precision-cut recesses (2×6.2 mm^2) spaced at 15 mm radial intervals. These slots securely house the $CaSO_4$:Dy TLD capsules at defined polar angles (30°, 60°, and 90°).

The experimental workflow involves pre-scanning the phantom via CT to verify the spatial geometry within a Treatment Planning System (TPS). The source is positioned at the geometric center, and an initial planned dose (e.g., 2 Gy at 15 mm) is prescribed to dictate the dwell time. Post-irradiation, TLDs mapped to specific (r, θ) coordinates are meticulously extracted and processed through a TL reader to acquire the raw dosimetric counts required for the TG-43 formalism equations.

4 Observation

The source geometry and setup parameters used for observation are:

$$L = 0.36\,\text{cm}, \quad r_o = 1.5\,\text{cm}, \quad \theta_o = 90^\circ$$

r(cm)	Angle (θ) (deg)	Counts 1	Counts 2	Average	β (Radians)	GL (r, θ) radians/cm2	gL(r)	F (r, θ)
1.5 cm	30	29904933	39569613	34737273	0.121157	0.448729	1.000000	11.0423
	60	32458400	41535246	36996823	0.207838	0.444426		11.8745
	90	36403253	38966076	**37684664.5**	2.902735	**5.375435**		1.0000
	120	40457142	45840920	43149031	0.207838	0.444426		13.8491
	150	39671953	37165631	38418792	0.121157	0.448729		12.2126
	210	45085594	36322026	40703810	0.121157	-0.448729		-12.9390
	240	38481899	41094967	39788433	0.207838	-0.444426		-12.7704
	270	33430040	34619312	34024676	2.902735	-5.375435		-0.9029
	300	34745531	38711851	36728691	0.207838	-0.444426		-11.7884
	330	35619607	36508503	36064055	0.121157	-0.448729		-11.4641
3.0 cm	30	9558770	15773545	12666157.5	0.060144	0.111378	0.557102	0.0461
	60	7865037	7601166	7733101.5	0.103923	0.111111		0.0281
	90	11734580	10120306	**10927443**	3.021736	**2.797904**		1.0000
	120	8632310	11221181	9926745.5	0.103923	0.111111		0.0361
	150	6590960	13539268	10065114	0.060144	0.111378		0.0367
	210	9505359	10505359	10005359	0.060144	-0.111378		-0.0364
	240	9468815	9935309	9702062	0.103923	-0.111111		-0.0353
	270	11127092	7780320	9453706	3.021736	-2.797904		-0.8651
	300	7218583	12367622	9793102.5	0.103923	-0.111111		-0.0356
	330	8200767	7791526	7996146.5	0.060144	-0.111378		-0.0291
4.5 cm	30	5084958	2545291	3815124.5	0.040043	0.049435	0.287951	39.1394
	60	4013754	2287266	3150510	0.069282	0.049383		32.3556
	90	3361639	4091248	**3726443.5**	3.061635	**1.889898**		1.0000
	120	2351444	5457893	3904668.5	0.069282	0.049383		40.1008
	150	3593595	2902154	3247874.5	0.040043	0.049435		33.3200
	210	4601623	3410326	4005974.5	0.040043	-0.049435		-41.0974
	240	2581057	4137905	3359481	0.069282	-0.049383		-34.5018
	270	2315239	2738344	2526791.5	3.061635	-1.889898		-0.6781
	300	3607983	3054696	3331339.5	0.069282	-0.049383		-34.2128
	330	7810408	5176991	6493699.5	0.040043	-0.049435		-66.6190

5 Conclusion

The observation table shows that the measured TLD counts decrease as the radial distance from the source increases. This is the expected physical behavior of brachytherapy photons, since the photon fluence is reduced by geometric spreading, attenuation in the phantom medium, and the finite source size. The highest response is obtained near the reference transverse plane, $\theta = 90^\circ$, where the path through the source encapsulation is minimal and the geometry is most favorable.

The variation of counts with angle also indicates the anisotropic nature of the source. Angles on opposite sides of the source show nearly symmetric behavior, while the response away from the transverse direction changes because of self-absorption inside the active core and filtering by the capsule. The sign change in β only reflects the orientation of the line joining the source end points to the point of interest; physically, the relevant quantity is the magnitude of the angular span. After geometry correction, the reference direction remains normalized to unity, confirming the TG-43 behavior of the source and validating the use of the line-source formalism for in-phantom dosimetry.

More generally, these dosimetric parameters are controlled by a combination of physical and geometric factors. The absorbed dose rate decreases with distance because of the inverse square law and additional attenuation in the phantom medium. The finite active length of the source introduces line-source geometry, so the dose depends not only on r but also on the polar angle θ through the angular span β. The encapsulation material, source self-absorption, and photon scatter in the surrounding medium all contribute to the radial dose function and anisotropy function. In the transverse plane, the geometry is most symmetric, so the measured response is used as the reference condition for normalizing the TG-43 parameters. These factors together explain the observed variation in TLD counts and the physical significance of the measured dosimetry trends.

References

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[3] M. J. Rivard, B. Coursey, L. A. DeWerd, W. F. Hanson, M. S. Huq, G. S. Ibbott, M. G. Mitch, R. Nath, and J. F. Williamson, "Update of aapm task group no. 43 report: A revised aapm protocol for brachytherapy dose calculations," *Medical physics*, vol. 31, no. 3, pp. 633–674, 2004.